“AccessPath Astana”

Diploma project proposal and design

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“AccessPath Astana” is a working product name. We propose a cross-platform mobile application for mobility-aware pedestrian routing in a limited Astana pilot area. This package defines the project, research basis, requirements, architecture, interface prototype, evaluation and delivery plan. The application and field dataset have not yet been implemented. The immediate decision is to review the selected pilot boundary, incorporate interview findings and begin field data auditing.

Evidence status: the literature and product comparisons use the sources listed in the references. Four interviews covering all four intended user groups were conducted on 17–22 October 2026 and are summarized from team notes in section 2. Participants agreed to anonymized use; the interviews were not recorded. These are self-reported experiences, not field-verified conditions. Supervisor approval of the general topic is reported by the team, while a documented review of these deliverables remains outstanding. Proposed thresholds and schedule targets are design decisions to validate, not measured results.

1. Project proposal

1.1 Project title and problem

Formal title: Development of a Cross Platform Mobile Application for Mobility Aware Pedestrian Routing in Astana.

The shortest pedestrian route may contain steps, unsuitable kerbs, steep segments, uneven surfaces or temporary obstructions. These characteristics affect people differently, so a generic walking route or a single accessibility label may be insufficient. Research supports personalized routing and careful handling of incomplete pedestrian data [R1–R8]. The extent of these problems in our pilot area has not yet been measured.

We will address a specific engineering problem: selecting a traversable route under a chosen set of mobility constraints while communicating gaps in the underlying data. The application will combine a pedestrian graph, profile-dependent rules, verified barrier reports and an accessible mobile interface. It will not certify that a route is universally accessible or that unobserved conditions are safe.

1.2 Users and team

Primary users are people using manual or powered wheelchairs, adults travelling with a child stroller, people using crutches, and older pedestrians who want to avoid particular physical obstacles. Four initial presets will provide an understandable starting point; users will review their own constraints because a group label does not determine individual capability. Secondary users are moderators and the project team collecting and validating data.

Team member	Main responsibility	Primary deliverables
Iskander Kustayev	Graph model and routing	Profiles, edge costs, Dijkstra and A*, report impact rules, algorithm tests, model and evaluation chapters
Alua Sagyndyk	Backend and geodata	REST API, PostgreSQL/PostGIS, import and audit pipeline, moderation, admin interface, Docker, data and architecture chapters
Altynay Bolatova	Mobile application and UX	Accessible mockups, map and route views, foreground GPS guidance, report submission, Android/iOS testing, interface chapter
All three	Research and integration	Interviews, literature, comparisons, requirements, integration tests, thesis and defense
Islam Omirzak	Academic supervision	Review of scope, methodological choices and key deliverables

1.3 Aim and objectives

Aim: develop and evaluate a cross-platform prototype that produces pedestrian routes satisfying explicit mobility constraints in a bounded area of Astana, using open geodata, field observations and moderated user reports.

O1. Identify and validate user needs through three to five real discussions and a structured analysis of relevant research.

O2. Build a versioned pedestrian dataset with documented provenance, missing attributes and a field-audited subset.

O3. Define four configurable mobility presets, hard exclusions, soft penalties and an explicit unknown-data policy.

O4. Implement Dijkstra and A* against the same weighted graph and verify equal optimal cost where their preconditions hold.

O5. Implement route, report and moderation APIs and a mobile client with map, route summary and basic foreground guidance.

O6. Evaluate route constraint compliance, data uncertainty, algorithm performance and completion of key user tasks against stated baselines.

1.4 Geographic and functional scope

The pilot is in Astana, within the area bounded by Turar Ryskulov Street, Orynbor Street, Kabanbay Batyr Avenue and Mangilik El Avenue. This replaces the earlier approximate EXPO and Botanical Garden landmark envelope. The four street boundaries are the team's agreed service

extent. A georeferenced GeoJSON polygon still needs to be produced from these boundaries and checked against the pedestrian network; no coordinates, area measurement or data completeness are asserted here. Import and endpoint validation will use the same approved polygon.

The initial audit will focus on a connected set of corridors and crossings serving approximately 20–30 evaluation origin-destination pairs. The exact sample will follow the data audit. A small import buffer may preserve graph connectivity; accepted routes must remain inside the agreed service polygon unless the user is explicitly told otherwise. We will not claim coverage of all of Yesil district.

Included in the diploma	**Excluded or conditional**
Four configurable mobility presets	Medical assessment or universal accessibility certification
Steps, kerbs, slope, surface and relevant width observations	Automatic recognition of all barriers from images
Moderated reports with photos and timestamps	Citywide live monitoring and automatic winter maintenance data
Android and iOS prototype testing	App store publication as a required result
Route map, textual instructions and foreground GPS	Voice navigation, background tracking and offline routing
Basic external lift connectors if verified data exist	Indoor routing and unverified live lift status
Controlled algorithm and route comparisons	Proven superiority over commercial services across Astana

1.5 Expected outcomes and engineering complexity

The final deliverables are a mobile prototype, backend API and moderator interface, a documented pilot dataset, a tested routing module, reproducible evaluation scripts and results, and a thesis with limitations. The engineering contribution is the integration of directional graph topology, per-profile edge eligibility, incomplete data, report lifecycles and explainable route output. Implementing two search algorithms alone is not the claimed novelty.

Success requires passing all mandatory acceptance criteria, reporting whether evaluation targets were achieved and explaining failures. The project can produce a valid research outcome even if some routes are unavailable because trustworthy accessibility data are missing.

2. Evidence of relevance

2.1 Evidence register

ID	Evidence or assumption	Status	Consequence
E1	Research describes heterogeneous mobility needs and incomplete accessibility data [R1–R8]	Published evidence outside Astana	Use configurable constraints and provenance
E2	Google Maps and 2GIS document stair avoidance [R9–R10]	Documented product capability	Avoid claiming that stair avoidance is unique
E3	Team selected the area bounded by Turar Ryskulov Street, Orynbor Street, Kabanbay Batyr Avenue and Mangilik El Avenue	Confirmed project decision	Encode and verify a single service polygon
E4	P01 and P02 describe limited acceptable detours; P03 and P04 favour more reliable routes	Interview self-report	Show distance and let users review individual constraints; do not infer universal detour limits
E5	P01, P02 and P04 express conditional willingness to report; P03 expects family assistance	Stated intention, not observed contribution	Keep reporting brief and moderated; do not assume equal participation
E6	P01–P04 describe different interaction or information needs	Interview self-report	Use explicit text, accessible controls and optional location access; validate the actual interface later
H2	Pilot OSM data lack some critical kerb and slope values	Unverified data hypothesis	Measure completeness before promising coverage
H4	Users understand the four presets and their settings	Interface usability hypothesis	Test wording and task completion; interview coverage alone does not validate the UI

2.2 User interviews

Four semi-structured interviews were conducted with representatives of the four intended user groups using the same 15-question guide. Participation was voluntary. Participants agreed to anonymized use of their answers; the team retained written notes only, with no audio or video recording. No diagnoses or direct personal identifiers are included in this report. Interviews were held in Russian; the accounts below are close summaries translated into English, not verbatim quotations. Participants are identified by ID only.

All four discussed experience in Astana. P02 also discussed the selected pilot area; specific incident locations have not been supplied and are not assigned to map features. The team introduced the project to participants. Whether participants subsequently checked the English summaries has not been established.

Interview P01

Participant ID: P01.
User group: parent with a child stroller.
Date: 22 October 2026.
Format: call.
Interviewer: Altynay Bolatova.

Answers: The participant walks daily with a stroller, mostly alone. On a recent difficult trip, a clinic had a high kerb with no ramp. She lifted the stroller while carrying a bag and holding the child, with help from a passer-by. A shop described as barrier-free had internal steps and no lift. She sometimes turns back at crossings without ramps. In winter, icy ramps and deep snow restrict outings. She uses 2GIS and a neighbourhood parents' chat, but reports not finding the ramp and step information she needs. She would accept a detour of approximately 200-300 metres for the described trips. She would report barriers if submission takes a few taps with a photo and the review outcome is visible. Small fonts, controls difficult to operate with one hand, continuous GPS tracking and unexpected route changes are unwelcome.

Findings: Kerbs without ramps, entrance steps, missing pavements and wide paving joints can interrupt stroller trips. A general accessibility label does not describe the entire journey. Detour tolerance depends on fatigue and the child. These are the participant's reported experiences, not verified attributes of particular pilot edges.

Requirement implications: Supports the stroller preset and configurable constraints (FR-01), step and kerb restrictions (FR-03), route distance (FR-05), brief photo-supported reporting and visible status (FR-08), and large controls (NFR-02). Explicit unknown-data wording and no silent rerouting strengthen FR-04 and FR-07. Rest places, feeding facilities and indoor shopping-centre lifts are recorded for future scope review.

Interview P02

Participant ID: P02.
User group: person using crutches temporarily.
Date: 19 October 2026.
Format: in person.
Interviewer: Alua Sagyndyk.

Answers: The participant has used crutches for approximately a month and travels by public transport to university and hospital. A recent crossing with steps was difficult because his hands were occupied and he could not reach his phone. At a building described as accessible, a steep, slippery ramp felt more difficult than stairs with a handrail. Slippery areas and uneven asphalt or paving are major concerns, including anticipated winter conditions. He reports that the navigation apps he uses do not provide the step information he needs. A detour of approximately 100-200 metres is acceptable for the described trips, while longer distances are tiring. He wants prominent

text describing unknown conditions rather than small icons. He would prefer one-tap or short voice reporting and spoken guidance that avoids looking at the screen. His discussion included the selected pilot area, without mapped incident coordinates.

Findings: Avoiding steps alone does not establish a suitable crutch route; slope, surface, fatigue and handrail availability also matter. Hands-busy interaction is a significant concern for this participant. His requirements may change as mobility changes.

Requirement implications: Supports editable crutch constraints, considering surface and slope alongside steps (FR-01 and FR-03), distance before starting (FR-05) and prominent warnings (FR-04 and NFR-02). Voice guidance, voice or one-tap reporting, handrail data and an automatically expiring injury setting are candidate extensions. They are not committed mandatory functions of the current pilot.

Interview P03

Participant ID: P03.
User group: older pedestrian who uses a cane.
Date: 20 October 2026.
Format: in person.
Interviewer: Iskander Kustayev.

Answers: The participant goes out daily, usually within his neighbourhood for shops or a clinic, and sometimes travels to his daughter by bus. High bus steps and pressure to board quickly made a recent trip difficult. At a clinic, a broken lift required him to climb stairs. He avoids steps without handrails and accepts a longer route, but needs places to sit. Fear of falling greatly reduces winter outings. He knows familiar routes and rarely uses a phone. He prefers large written warnings because colours and icons are difficult to distinguish. He is unlikely to submit barrier reports independently and would expect family help. Small text, many taps, tracking and automatic changes are unwelcome. A large button to contact relatives or a volunteer would be useful to him.

Findings: Fear of falling, rest needs and difficulties at transport or building interfaces affect this participant's mobility. A map-and-reporting application may not be used independently by every intended user. Bus boarding and indoor lift incidents motivate the problem but are outside the outdoor pilot's routing responsibility.

Requirement implications: Supports the older-pedestrian preset (FR-01), a simple interface with large text and non-colour warnings (NFR-02), optional GPS access (FR-02) and no silent route changes (FR-07). Rest-place data and a help button are future candidates. Reporting uptake among older users must not be assumed. A preference for displaying a last-check date is not explicitly established in this summary.

Interview P04

Participant ID: P04.
User group: manual wheelchair user.
Date: 17 October 2026.
Format: call.
Interviewer: Altynay Bolatova.

Answers: The participant has used a manual wheelchair for several years and travels independently around the city every day. On a recent trip to a cafe described as accessible, a high kerb without a ramp left him waiting for help. He identifies kerbs, steep slopes, poor surfaces, narrow passages and parked cars as recurring problems. Thresholds, narrow doors and non-working lifts can also prevent access on arrival, so a single accessibility label is insufficient. Winter is particularly difficult. He uses 2GIS and Google Maps, photographs, reviews and a friends' chat. He wants kerb height, slope, width, surface, lift status and the date of the last check. He accepts a substantial detour if it is more reliable, describing an extra ten minutes as preferable to getting stuck. He wants unknown segments clearly identified so that he can make an informed decision. He would submit type, location and photo if reports are checked. Small buttons, a jumping map and retention of route history are unwelcome.

Findings: Vague accessibility labels can omit journey-critical details. This participant values measurable attributes, provenance and freshness, and wants control over individual constraints. Interview statements about lift status do not establish that verified live lift data exist for the pilot.

Requirement implications: Supports editable wheelchair constraints (FR-01), relevant kerb, slope, width and surface attributes (FR-03), explicit unknowns (FR-04), short reports and visible review status (FR-08), provenance and observation dates (FR-14), accessible controls (NFR-02) and minimal location retention (NFR-05). External lifts remain conditional (FR-15). Peer confirmation is a possible design approach, not an explicit request established by the supplied answer summary; moderator verification remains the baseline.

2.3 Cross interview analysis and requirement decisions

All four summaries describe steps or kerbs that complicate travel and winter-related concerns. Surface difficulties are explicit in P01, P02 and P04; P03 emphasizes even routes and fear of falling. P01, P02 and P04 describe a place whose accessibility description did not match their experience. P03 describes a broken lift, without stating that the location was labelled accessible. Numeric limits and attribute completeness therefore remain more informative design targets than a single yes/no label.

Detour preferences differ: P01 describes approximately 200-300 metres, P02 approximately 100-200 metres, and P03-P04 prioritize more reliable routes without a common distance limit. These are contextual preferences, not calibrated universal thresholds. The initial profiles remain editable; numeric slope, kerb and penalty values require further justification and field evaluation.

All four raise interaction difficulties. Tracking concerns are explicit in P01, P03 and P04; P02 emphasizes hands-busy operation. P01 and P03 object to automatic changes, while P04 objects to an unstable map. The design therefore retains optional GPS, minimal data retention, clear warnings and user-controlled route rebuilding. P04 explicitly requests the last-check date; the same preference is not directly established for every participant. Date display is also supported by the literature and provenance design.

P01, P02 and P04 express conditional willingness to report, with differing effort limits. P03 expects help from family. This supports brief reporting and visible moderation, but does not establish actual participation rates. P04 wants to decide about unknown sections, while P03 favours more predictable paths. Strict routing remains the default; exploratory routing requires explicit selection and never overrides a known prohibition.

Finding and evidence	Requirement decision	Scope and verification
Different detour preferences and mobility constraints, P01–P04	Retain editable presets, hard restrictions and visible distance; FR-01, FR-03, FR-05	Core; T01, T03, T05; no universal detour limit inferred
Missing information and unreliable labels, P01–P04; freshness request, P04	Explicit unknowns and barrier provenance; FR-04, FR-14	Core; T04, T14; observed dates shown as unknown when unavailable
Brief reporting and review visibility, P01, P02, P04; assisted reporting, P03	Streamlined photo-supported form and own-report status; FR-08	Core; T08; observed usability and uptake remain untested
Hands-busy use and readability, P01–P04	Large controls, prominent text and stable interaction; NFR-02, FR-07	Core; T07, T17; voice features deferred for feasibility review
Tracking and unwanted changes, P01, P03, P04	Optional GPS, explicit rebuild and minimal location retention; FR-02, FR-07, NFR-05	Core; T02, T07, T20
Rest places, handrails, help and voice interaction	Record candidate extensions rather than expanding the mandatory pilot	Future backlog; no acceptance or delivery claim

The interviews do not establish that Google Maps or 2GIS lack stair-avoidance functionality. Participants describe the information available to them in their own use. The official capability comparison in section 4 remains separate from these reported experiences.

2.4 Method limitations and evidence records

Four participants cover the intended user groups in Astana, but this is a small requirements-exploration sample. Accounts are self-reported and were not independently checked in the field. P02's pilot-area familiarity does not establish coverage of all paths within the boundary. Recruitment details, interview durations and participant checking of the translated summaries have not been supplied; they are not assumed. The project was introduced to participants, which does not itself validate the wording of these summaries or the finished interface. Findings cannot support population percentages, demand forecasts, algorithm performance or achieved accessibility claims.

Retain the anonymized session notes, the actual 15-question guide and the consent record with appropriate access. The report contains translated summaries rather than full question-by-question transcripts. Requirements should cite P01-P04 only where the supplied accounts support the decision. Implementation feasibility and usability must be tested separately.

3. Literature review

3.1 Selection and appraisal

The review uses eight scientific publications [R1–R8] and supporting primary documentation [R9–R25]. Seven of the eight scientific works were published in 2022–2026; the 2019 Project Sidewalk study is retained as a foundational deployment study. Searches on 20 October 2026 covered personalized accessible routing, wheelchair routing and OSM, sidewalk data quality, crowdsourced barriers and user perceptions. This is a focused review, not a systematic review or a claimed Scopus search.

Primary publisher pages, author repositories and official documentation were checked. For R1, R2 and R3 the appraisal is limited to accessible publication records and abstracts; numerical experimental claims are not inferred from unavailable full text. R4–R8 were examined through available article text or author versions. Limitations below include our appraisal of transferability, not necessarily claims made by each author.

Source	Method and evidence	Main contribution	Limitation and project use
R1 Majic et al 2026	OSM suitability analysis and wheelchair network case study in Graz	Identifies relevant tags and compares pedestrian and wheelchair routes	Another city; local completeness is unproven. Audit pilot tags and compare identical OD pairs
R2 Nguyen et al 2024	MyPath combines smartphone surface sensing with accessible route generation	Connects surface information, OSM and routing	Sensor collection exceeds our scope. Use manual surface observations; retain surface-aware cost
R3 Darvishy et al 2022	Zurich case study and accessible-routing vision	Identifies required attributes and combined collection methods	A vision and local study do not establish Astana coverage. Combine import with field audit
R4 Darko et al 2022	Adaptive preference model with reinforcement learning	Treats changing preferences and sidewalk conditions in route choice	Model and evaluation differ from static search. Keep explainable fixed snapshots and defer adaptation
R5 Labbé et al 2023	Workshops with 51 stakeholders	Highlights trust, usability and uses of accessibility data	Qualitative context is not local demand evidence. Design provenance and real consultation
R6 Askari et al 2025	Compares virtual audits with government field data in two US settings	Finds substantial agreement for several infrastructure features	Severity definitions complicate comparison. Use an explicit field-audit rubric

R7 Li et al 2025	Survey of 190 people from five mobility-device groups using 52 barrier images	Shows differences and shared concerns across groups	Image judgments do not establish actual route traversability. Keep profile settings configurable
R8 Saha et al 2019	18-month remote crowdsourcing deployment and quality analysis	Demonstrates scalable sidewalk labelling and quality trade-offs	Imagery and crowd labels require validation. Moderate reports and retain evidence

3.2 Supporting source analysis

Source	Role in this project	Limitation
R9 Google Maps official announcement	Documents stair-free walking routes and data-dependent availability	Does not validate the selected Astana corridors
R10 2GIS official help	Documents walking routes, crossings, gates and stair exclusion	Local route quality still requires testing
R11 Wheelmap official description	Demonstrates place-level accessibility ratings and unknown status	Place ratings alone do not prove continuous path accessibility
R12 openrouteservice documentation	Shows configurable wheelchair restrictions	Capability depends on underlying data and deployment
R13 and R14 OSM tagging documentation	Defines slope and kerb representation	Tag presence and correctness are not guaranteed
R15 WCAG and R16 Flutter guidance	Inform contrast, labels and mobile accessibility testing	Checklist compliance is not a substitute for user testing

3.3 Critical synthesis and project gap

The reviewed routing work supports personalization, but approaches differ in complexity and evidence. A richer sensing or adaptive model can capture more conditions, while increasing data and validation requirements. For a three-person diploma, a transparent constrained graph model is a defensible starting point. We will state its limits: preferences remain fixed within one request, effort is approximated rather than physiologically measured, and uncertain data can reduce route availability.

The data studies indicate that collecting labels at scale and trusting every label are separate problems. Our design therefore stores source, observation time, verification status and unknown attributes. The engineering implication is that a route result must describe both the path and the

evidence behind it. A short route with missing kerb data cannot receive the same explanation as a field-checked route.

The human-centred studies support configurable profiles and consultation. The four Astana interviews add local self-reported examples of barriers and differing detour, interaction and reporting preferences. They inform the decisions in section 2.3 but do not validate the finished interface, calibrated routing parameters, adoption or population demand. Field auditing and later usability evaluation remain necessary.

The proposed contribution is a locally evaluated integration of four mobility presets, a versioned pilot graph, moderated temporal reports and uncertainty-aware route explanations. We do not claim to invent accessible routing or to be the first such solution in Kazakhstan. The research question is whether this bounded implementation reduces known constraint violations relative to distance-only routing while keeping detours and computation cost explicit.

4. Existing system comparison

We compare four relevant systems across the same eight criteria. Google Maps and 2GIS are general navigation comparators; Wheelmap is an accessibility information comparator; openrouteservice is a technical accessible-routing comparator. This distinction prevents treating a place-rating service as though it promised a complete navigation workflow.

Legend: D - documented in the cited source; U - not established by the reviewed material, not proof of absence; P - proposed for AccessPath, not implemented. No hands-on Astana route comparison has yet been conducted.

Criterion	**Google Maps R9**	**2GIS R10**	**Wheelmap R11**	**openrouteservice R12**
Main function	D navigation	D navigation	D place accessibility	D route engine/API
Pedestrian route generation	D	D	U	D
Stair avoidance	D where data available	D filter	U at route level	D route options
User-set slope and kerb limits	U	U	U	D wheelchair restrictions
Surface or smoothness restrictions	U	U	U at route level	D
Four proposed presets	U	U	U	U as this exact set
Accessibility contributions	D place attributes in R9	U in reviewed page	D place ratings	OSM-based; app workflow U
Explicit unknown information	Local attribute coverage U	Local attribute coverage U	D unknown place rating	Attribute dependent; UI U

For AccessPath all eight criteria are design targets, except that the main function is a bounded research prototype. Planned differences are explicit segment uncertainty, four reviewable presets and a documented report lifecycle linked to a local field-audited graph. These are requirements, not achieved advantages. In particular, stair avoidance is already available in both general navigation comparators.

Before the final defense, record product version, platform, date, selected route options, identical endpoints, route availability and permitted screenshots or manual observations. Keep unsupported cells marked U until evidence changes them. Do not rank a product poorly because a feature was not mentioned in a single help page. Route-level evaluation must distinguish missing service coverage from an actual failed route.

5. Technology comparison

The following choices are proposed engineering decisions for this project. Documentation supports capabilities; suitability judgments reflect our scope and must be revisited after an initial integration spike. Versions will be pinned at implementation rather than claiming that the newest release is required.

5.1 Server platform

Alternative	Fit to requirements	Cost or limitation	Decision
Python with FastAPI R17	Typed request validation, OpenAPI, direct integration with Python graph code	CPU-heavy routing needs bounded workers and measurement	Selected for the pilot
TypeScript with NestJS R18	Structured modules and a familiar JavaScript ecosystem	Python routing would add a language boundary, or need reimplementation	Viable alternative
Java with Spring Boot R19	Structured application and strong backend tooling	More setup for this team's small Python-based evaluation workflow	Reserve alternative

The backend will be a modular monolith with an internal routing package. This reduces deployment interfaces while retaining clear component boundaries. A separate routing service becomes justified only if measured load or independent deployment requires it.

5.2 Data storage

Alternative	Fit to requirements	Cost or limitation	Decision
PostgreSQL with PostGIS R20	Relational reports and audit records plus spatial distance queries	Server administration and migration discipline	Selected
SQLite with SpatiaLite R21	Spatial SQL in a portable local file	Less aligned with concurrent server moderation and shared writes	Useful for local experiments
MongoDB R22	GeoJSON and spatial querying with flexible documents	Relationship and validation rules would need a different modelling approach	Viable but not preferred

PostGIS stores authoritative geometries; routing uses an in-memory adjacency representation derived from a versioned snapshot. Database spatial proximity does not by itself establish that two paths are connected.

5.3 Mobile platform

Alternative	Fit to requirements	Cost or limitation	Decision
Flutter R16	Shared Android/iOS interface and accessibility testing support	Team must learn Dart and verify native permissions	Selected
React Native R23	Shared code and platform accessibility APIs	Platform-specific behaviour still needs testing	Viable alternative

Use flutter_map for map display [R24], with a configured tile source and visible attribution. The map widget does not perform our accessible routing. Device GPS and camera integrations require an early Android/iOS proof of concept. Provider terms and available tile access must be checked before deployment; bulk offline tile downloads are outside scope.

5.4 Component interaction

Alternative	Fit to requirements	Cost or limitation	Decision
HTTPS REST with JSON and OpenAPI R17	Simple route/report operations, inspectable contracts and mobile integration	More verbose payloads	Selected
gRPC R25	Typed RPC with generated clients	Additional tooling without an established pilot need	Reserve for later service separation

Photos use multipart upload and private file storage on a backed-up server volume for the pilot. PostgreSQL stores file metadata and access permissions. Docker Compose is the proposed reproducible deployment method. Remote deployment is not established in this package; the repository URL must be added when available.

6. Requirements specification

Must means required for the agreed final prototype. Should means desirable if the core requirements pass. All non-functional requirements are Must. The requirements incorporate the interview decisions in section 2.3. Interview support does not imply implementation or usability acceptance; further evaluation may revise them. Test IDs in section 9 provide traceability.

6.1 Functional requirements

ID and priority	Requirement	Acceptance criterion
FR-01 Must	Select and review four presets	All four presets are available and editable; reviewed constraints are saved and sent unchanged in a route request; T01
FR-02 Must	Select origin and destination	Manual selection works without GPS; outside-area and unconnected points receive explicit errors; T02
FR-03 Must	Enforce profile restrictions	Every returned route contains zero edges violating known hard constraints in fixtures; T03
FR-04 Must	Handle unknown critical attributes	Strict mode excludes unknown required attributes; exploratory mode requires explicit selection and shows warnings; T04
FR-05 Must	Explain route result	Distance, cost meaning, unknown-length share, snapshot ID and relevant warnings are shown; T05
FR-06 Must	Handle no suitable route	Disconnected or blocked fixtures produce a no-route result without silently relaxing restrictions; T06
FR-07 Must	Basic foreground guidance	With permission, position and next textual instruction update on a replayed GPS trace; inaccurate fixes trigger a warning; rebuilding requires an explicit action and never silently changes constraints; T07
FR-08 Must	Submit a barrier report	Type, point, observation time and description validate; optional supported photo uploads; receipt confirms pending status; contributors can view their own pending, confirmed or rejected status; T08

FR-09 Must	Provide low-friction access	Browsing and routing require no account; report submission requires an authenticated contributor; T09
FR-10 Must	Moderate and audit reports	Only moderators may confirm, reject, resolve or mark stale; actor, time and reason are retained; T10
FR-11 Must	Apply verified report changes	A new route request uses the committed report version; an already displayed route is labelled with its version and can be refreshed; T11
FR-12 Must	Preserve lifecycle and uncertainty	Temporary stale reports become unresolved information; permanent barriers are not cleared merely by age; T12
FR-13 Must	Import and inspect geodata	Import creates versioned nodes/edges and a completeness summary; malformed records are flagged; T13
FR-14 Must	Show barrier details	Barrier card shows source, verification status and observation and verification dates without publishing contributor identity; missing dates are explicitly unknown; T14
FR-15 Should	Verified external lift connectors	A connector is eligible only under documented profile and status rules; unknown operation is excluded in strict mode; T15

6.2 Non functional requirements

ID	**Proposed requirement and testable target**
NFR-01 Performance	Route API p95 at most 2 seconds for a warmed graph of at most 50,000 directed edges, 10 concurrent clients and a declared 2-vCPU/4-GB test server; network time reported separately; T16
NFR-02 Interaction	Touch targets at least 48 logical pixels; ordinary text contrast at least 4.5:1; 200% text scaling preserves key tasks; status has text and shape as well as colour; one-handed report entry is included in usability checks; T17 [R15–R16]
NFR-03 Platform and robustness	Core tasks pass on one recorded Android device and one iOS device or simulator; denied GPS/camera permissions and network errors have recoverable states; T18

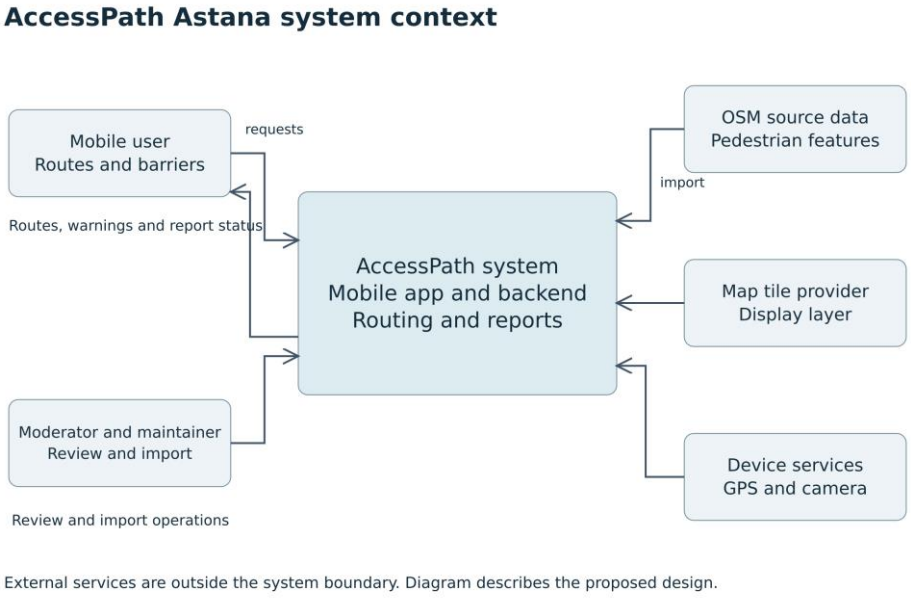
NFR-04 Reproducibility	Every benchmark records graph/report/profile versions, random seed, endpoints, algorithm and environment; reruns reproduce path cost within tolerance; T19
NFR-05 Access and data minimization	Moderator mutations reject unauthorized requests; HTTPS for deployed traffic; photos remove EXIF; no continuous location history stored by default; T20
NFR-06 Upload integrity	Accept only validated JPEG/PNG up to 5 MB; reject malformed uploads and duplicate submissions using an idempotency key; T21
NFR-07 Test quality	All mandatory scenario tests pass; every known hard exclusion has a positive and negative fixture; failures are recorded rather than hidden by coverage percentage; T22
NFR-08 Recovery	Restore the database and photo volume from a test backup and verify report/photo consistency; document observed recovery time; T23

The two-second target and graph limit are proposed acceptance targets, not benchmarks already achieved. Accessibility values are project targets informed by guidance; passing selected checks is not a claim of complete WCAG certification.

7. Architecture and diagrams

7.1 System context

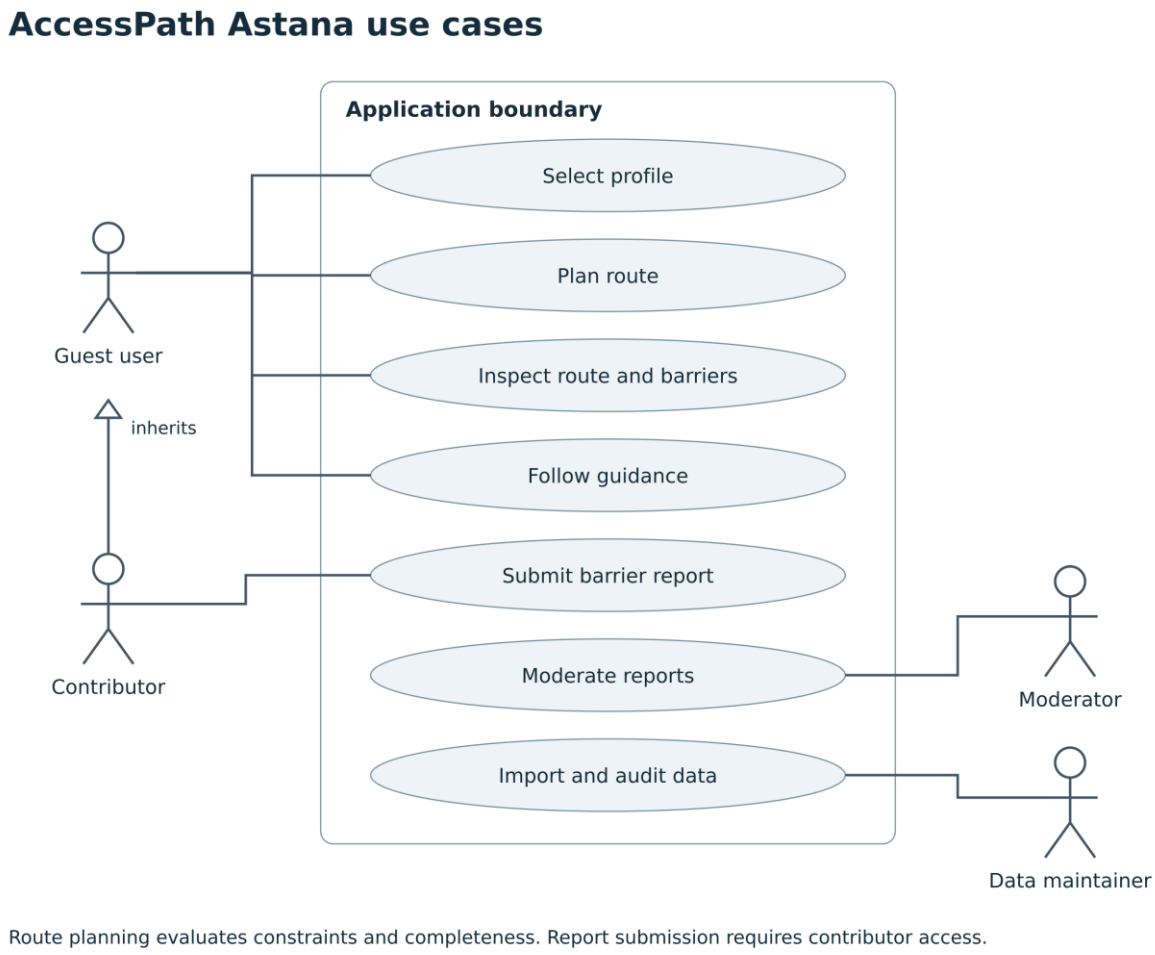
The mobile user selects constraints and requests routes; the contributor submits evidence; the moderator reviews reports; and the data maintainer imports and audits pedestrian data. External dependencies are the OSM dataset, a tile provider and device location/camera services. The system boundary includes the mobile client, backend, routing package, database, media storage and moderator interface.



System context diagram

7.2 Use cases

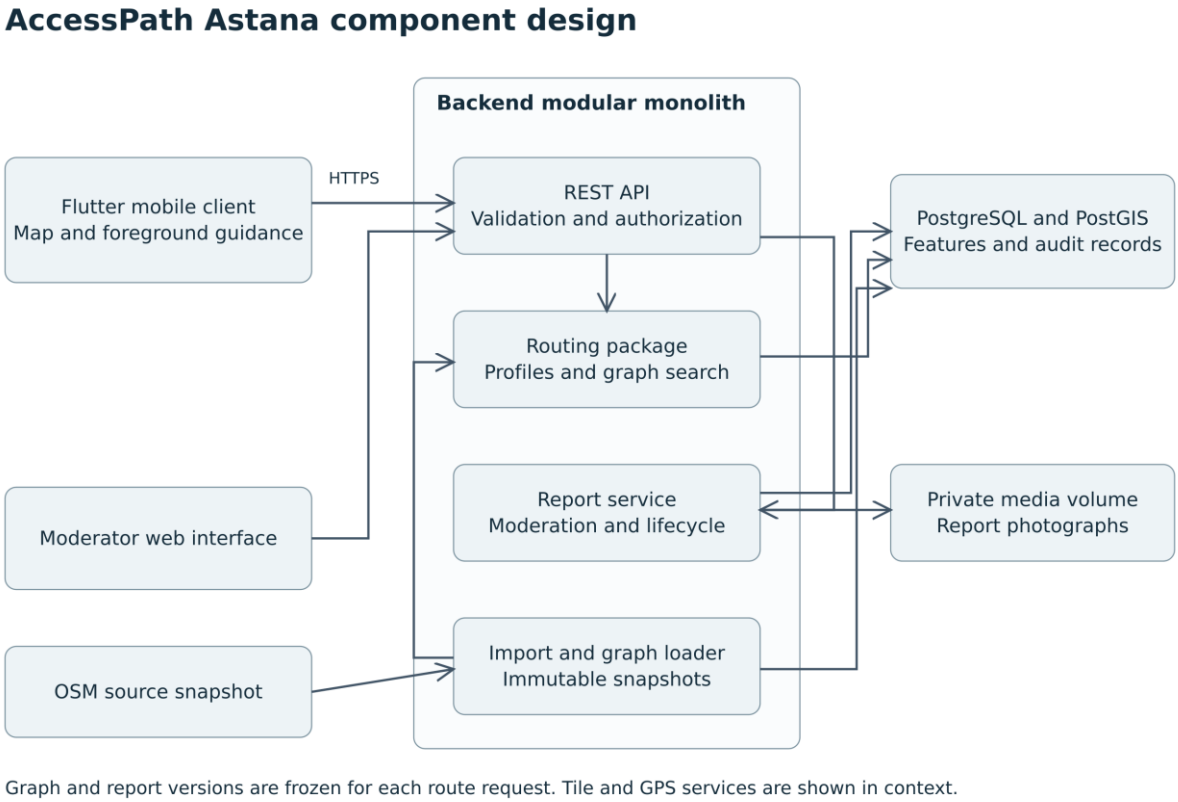
Guest users can select a profile, plan a route, inspect barriers and use guidance. Contributors inherit these capabilities and can submit reports. Moderators review reports; data maintainers import and inspect data. Route planning always evaluates profile constraints and data completeness. Sending a report does not automatically confirm it.



Use case diagram

7.3 Component design

The API validates requests and delegates to routing or report services. The graph loader builds immutable snapshots from the database. Moderation commits a report version and triggers a snapshot refresh. Routing returns the graph and report versions used, avoiding mixed versions within one request. The mobile client receives geometry, warnings and instructions through the same contract.



Component diagram

Editable SVG sources and Mermaid or PlantUML text are supplied alongside readable PNG exports. This is a proposed architecture, not an implemented deployment.

7.4 Data model and collection

Entity	Key fields and relationships
PilotArea	ID, agreed polygon, version, approval date
Node	ID, point geometry, source reference, level or crossing context
Edge	From/to nodes, geometry, length, direction, slope, surface, width, steps and crossing/kerb references
Observation	Feature ID, attribute, value and unit, source, method, observed_at, verification status

MobilityProfile	ID/version, hard limits, penalty weights and unknown policy
BarrierReport	Contributor, point, candidate edge, type, evidence, timestamps and lifecycle status
ModerationAction	Report ID, moderator, previous/new status, reason and timestamp
Photo	Report ID, private storage key, media type, size and stripped-metadata status
GraphSnapshot	Import version, observations version, report version, build timestamp and checksum

One report may be associated with multiple candidate edges until reviewed. At crossings, distinguish the two kerbs and travel directions. Crossing lines on a map are not automatically connected: bridge, tunnel, level and crossing information must be inspected. Unknown values remain null with an explicit status; they must not become zeros or 'accessible'.

Import OSM paths and tags, retain provenance and generate a missingness report. Walk a selected audit subset using a consistent form. Record kerb measurements in metres and slope as a directional percentage where measurement is defensible [R13–R14]. Photograph only relevant infrastructure. If equipment cannot measure a feature reliably, mark it unknown. Keep original data and manual observations separately so later imports do not erase verified changes.

7.5 Graph and routing model

Use a directed graph G = (V, E). Split edges at junctions and relevant barriers. Origin/destination snapping must respect actual pedestrian connectivity: a nearby path across a fence is not automatically reachable. A failed connection returns a location error rather than an invented crossing.

For profile p, first remove edges violating known hard constraints. In strict mode also remove edges missing any attribute required to establish eligibility for that profile. In exploratory mode permit selected unknown attributes with a nonnegative penalty and explicit warnings. Neither mode overrides a known prohibition.

For remaining edges use $C(e,p) = L(e) \times [1 + ws(p) \times S(e) + wr(p) \times R(e) + wu(p) \times U(e)] + B(e,p)$. L is geometric length in metres; S, R and U are normalized nonnegative slope, surface and unknown indicators. Weights are nonnegative. B is a nonnegative barrier penalty in equivalent metres; an impassable barrier is excluded, not merely penalized. This is generalized cost, not travel time. Thresholds and weights will be calibrated using consultation and field fixtures and versioned before evaluation.

A* uses planar straight-line distance to the target in the same local metric coordinate system used for edge lengths. Because edge cost is at least edge length and length is at least endpoint displacement, this heuristic is a lower bound and is consistent. Validate those assumptions during graph construction, including lift connectors. Use $h = 0$ if they cannot be guaranteed. Dijkstra and A* share eligibility rules, costs, graph snapshot and stopping rules.

Initial preset behaviour: wheelchair and stroller presets exclude known steps; crutch and older-pedestrian presets initially favour step avoidance and can be adjusted before routing. All presets expose distance trade-offs and unknown policy. Numeric slope and kerb limits are configuration values to validate, not medical advice or universal standards. Benchmarks must not mix different preset versions.

7.6 Report trust and temporal handling

Reports progress from pending to confirmed or rejected; confirmed reports may become resolved or stale. Only confirmed reports alter eligibility, while pending reports can be displayed as unverified information. A moderator must inspect location, evidence and conflicts before confirmation. Duplicate reports do not create independent votes automatically.

For temporary information, an exploratory confidence indicator may use $q(t) = q0 \times \exp(-age/\tau)$, where q0 reflects evidence class and τ depends on barrier type. This is a proposed prioritization score, not a calibrated probability of safety. It schedules rechecks and affects warnings; it must not automatically reopen a confirmed blocked edge. A stale unresolved closure remains excluded in strict mode until rechecked. Permanent steps and kerbs do not disappear through time decay. Cold-start operation uses moderator review rather than an unvalidated reputation score.

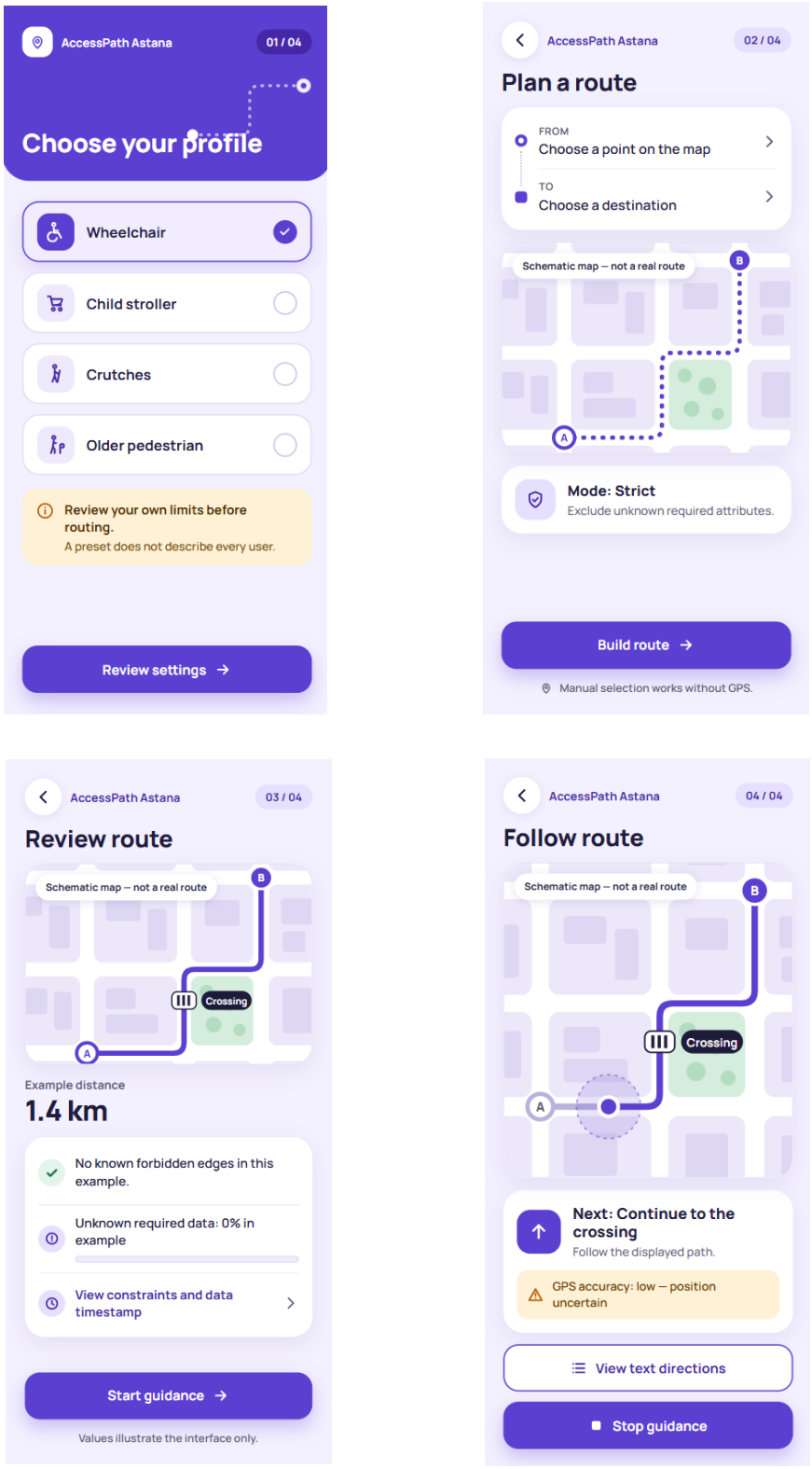
7.7 API contract outline

Endpoint	**Contract and outcome**
GET /v1/profiles	Versioned presets and editable settings
POST /v1/routes	Origin/destination, profile version, overrides and unknown policy; returns GeoJSON line, instructions, distance, warnings and snapshot IDs
GET /v1/barriers	Bounding-box query; returns public barrier details without contributor identity
POST /v1/reports	Authenticated multipart report; returns report ID and pending status
GET /v1/reports/{id}	Contributor's own report or moderator view
PATCH /v1/moderation/reports/{id}	Authorized state change with reason and expected version
POST /v1/admin/imports	Authorized import job for a declared polygon and source snapshot

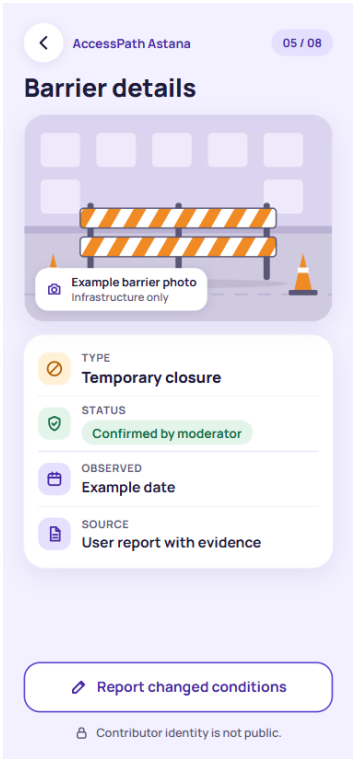
Use longitude/latitude order for GeoJSON. Validate coordinate ranges, pilot membership and timestamp formats. Distinguish OUTSIDE_PILOT, NO_CONNECTED_EDGE, NO_ROUTE and DATA_INSUFFICIENT. A no-route outcome is not a server crash. Concurrent moderation must reject stale versions instead of silently overwriting another review.

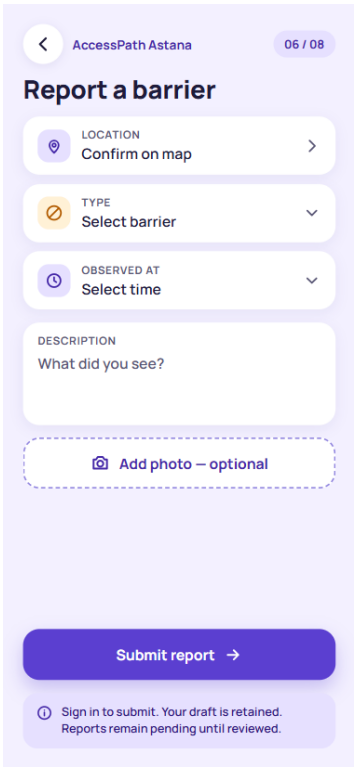
8. UX and prototype

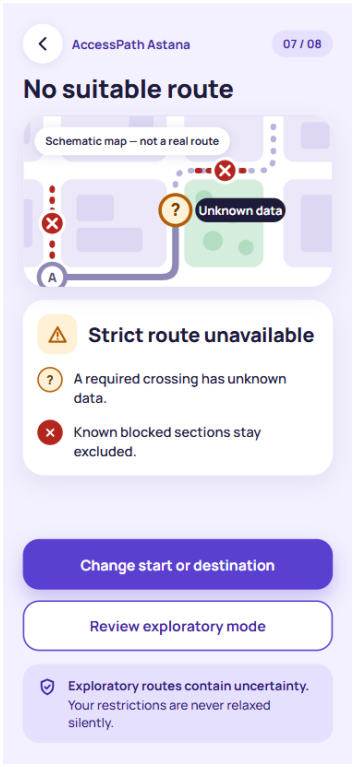
The supplied mockups are design prototypes with illustrative values and a schematic map, not measured routes. They cover profile selection, route planning, route review, guidance, barrier details, reporting, no-route feedback and moderation. SVG sources can be imported into a design tool and edited.

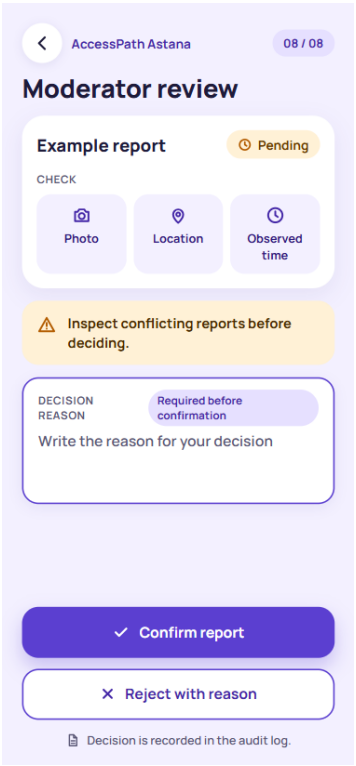


Prototype screens one to four









Prototype screens five to eight

Scenario U1: choose a wheelchair preset, review constraints, select two pilot points, build a strict route, inspect distance and uncertainty, then start guidance. If strict routing fails, explain missing or blocked information; offer changing endpoints or explicitly selecting exploratory mode without implying that it is verified.

Scenario U2: inspect a barrier card, choose report, confirm its location, select type, add description and optional photo, and submit. Show a pending receipt, provide access to the contributor's report status and explain that review is required before the report changes routing. If the user is not signed in, preserve the draft through sign-in.

Scenario U3: a moderator reviews evidence and location, checks conflicting observations, makes a decision with a reason and verifies the resulting route snapshot. The confirmation action has a clear consequence and cannot be triggered by an ordinary contributor.

Scenario U4: a user denies location permission and still selects a route manually. Guidance indicates that live position is unavailable. If GPS accuracy is poor, retain the route and avoid claiming a precise current step. Off-route detection offers a rebuild action rather than silently changing constraints.

Use large labelled controls, readable route text and non-colour status markers. Provide a textual route list as an alternative to the map. Review key screens at 200% text scaling and with TalkBack/VoiceOver. The mockups specify interaction intent; implementation and task-based user testing remain future work. Interviews support large controls, explicit uncertainty, a stable map and user-controlled route rebuilding. Voice guidance, voice reporting, benches, handrails, a help button and automatic profile expiry are future candidates, not functions demonstrated by these screens.

9. Verification and evaluation plan

9.1 Test mapping

Test IDs	Method	Evidence to retain
T01–T06	Unit and API fixtures for profiles, snapping, exclusions, unknowns, explanations and no-route cases	Fixture graph, expected result and automated log
T07	Replay recorded or synthetic GPS traces with permission and accuracy variations	Trace, instruction sequence and screen recording
T08–T12	Report lifecycle and authorization integration tests	API responses, audit log and route version change
T13–T15	Import validation, barrier details and conditional lift fixtures	Import report and source/observation comparisons
T16	Load test under declared server and graph conditions	Raw timings, concurrency configuration and p95
T17–T18	Accessibility and cross-platform scenario checks	Device/OS list, checklist and defects
T19	Reproducibility rerun	Manifest, seeds and cost comparison
T20–T21	Access control, image validation and metadata inspection	Rejected requests and stored-file inspection
T22–T23	Mandatory suite and restoration rehearsal	Test summary and restore record

Unit fixtures must cover a unique path, equal-cost alternatives, an unreachable goal, a directed slope, steps, kerb limits, unknown critical attributes, a confirmed closure and origin/destination splitting. Include adversarial topology such as crossing geometries at different levels. Test floating-point costs with an explicit tolerance; tied optimal paths may have different geometry.

9.2 Three separate comparisons

Algorithm baseline: Dijkstra against A* using exactly the same eligible graph, weights and request set. Primary correctness criterion is equal cost within $1e\text{-}6 \times \max(1, \text{cost})$. Compare search time and vertices expanded; a speed-up is a hypothesis, not a required positive result.

Route-quality baseline: distance-only shortest path on the same pedestrian topology, without mobility-specific exclusions. Evaluate both outputs against the same independently audited constraints. Also report the shortest eligible route as a second reference for the cost of soft preferences. If the accessible system returns no route, count that explicitly; do not omit failed requests from the denominator.

External comparison: Google Maps and 2GIS on identical endpoints, dates and their available accessibility options. Since their graphs and costs are unknown, this is a descriptive comparison,

not a controlled algorithm benchmark. Record unavailable service or absent output separately. Do not assert a win where route conditions have not been checked.

9.3 Measurement design

Select 20-30 origin-destination pairs spanning crossings, alternative paths, different surface conditions and incomplete data; apply all four frozen profiles. Reserve a separate set of at least ten pairs for tuning, where feasible, and do not tune on the final evaluation pairs. The actual sample and exclusions will be documented after the pilot audit.

Freeze graph, reports and profile configuration for each run. Record CPU, memory, OS, code commit, graph size, connected components and missing attribute rates. Use five warm-up runs and 30 timed repetitions per request per algorithm, randomizing order. Report median and p95, expanded vertices, route cost, no-route rate and cold-load time separately. Pair outcomes by request and avoid treating repeated timings as independent users.

Measure field agreement on an independently rechecked subset: a second team member audits the features without seeing the algorithm's recommendation, then disagreements are adjudicated. Define a barrier violation per traversed feature under a profile. Report both route-level violation rate and unknown-length share. A target of zero known hard violations is mandatory; a lower violation rate than the distance baseline is an evaluation hypothesis. Do not infer actual-world zero risk from zero fixture violations.

Detour ratio = accessible route length / distance-only route length, where both exist. Unknown-length share = length of segments missing at least one required attribute / total route length, counting each segment once. Report coverage and no-route rates beside detour statistics to avoid rewarding a system that simply refuses most requests.

Conduct formative usability tests with approximately five to eight volunteers if recruitment permits. Ask them to select a profile, plan a route, interpret uncertainty and submit a report in the prototype or app. Record task completion, errors, assistance and comments. A provisional target is at least 80% completion of core tasks without assistance; with a small sample report counts and avoid population claims. The requirements interviews in section 2 are not task-based usability tests; no actual usability results are available yet. Include hands-busy or one-handed operation and checking report status among the evaluated tasks.

9.4 Threats to validity

The pilot may not represent all Astana neighbourhoods. Data collected in one season cannot establish year-round accessibility. Convenience recruitment may miss important mobility needs. GPS errors, observer disagreement and stale barriers may affect outcomes. Product versions and provider data can change. We address these by publishing the evaluation envelope, dates, uncertainty, sampling procedure and failure cases. Weather prediction and automatic ice detection are outside scope; recent reports may describe conditions without guaranteeing completeness.

10. Project management and risks

10.1 Repository and task workflow

The source package supplies documentation, editable diagrams, mockups, a README, task register and PR template. The team reports that YouTrack has been created and tasks assigned; its URL and actual task IDs have not yet been supplied for this package. The package itself is not a hosted GitHub/GitLab repository. Add the actual service links, verify access for each member and Islam Omirzak, and grant committee access before submission. Each member must use an individual account.

Create one task per meaningful deliverable with owner, priority, expected outcome, acceptance criteria and deadline. Use a branch such as AP-06-routing-model and reference the actual task key in commits and PRs. Proposed AP identifiers in the task register must be mapped to real YouTrack IDs when created. Review changes before merging into main; attach the PR link to the task and mark it complete only after acceptance.

The initial documentation package may be committed transparently as a starting draft. Subsequent research, corrections and design decisions should be contributed progressively by their actual authors. Do not backdate commits or attribute generated material to work that teammates have not reviewed. Tag the exact first submission pre-defense-1 only when ready; never move a submitted tag. Provide completed work, limitations and outstanding tasks with the tag.

10.2 Initial task register

Proposed task	**Owner**	**Due date**	**Acceptance outcome**
AP-01 Repository and YouTrack setup	Iskander	24 Oct 2026	YouTrack created and tasks assigned per team; add actual URLs and verify repository/access
AP-02 Pilot boundary and feasibility audit	Alua	24 Oct 2026	Street boundaries agreed; encode polygon and audit data
AP-03 User consultation records	Altynay	24 Oct 2026	Four dated interviews summarized; retain notes, guide and consent record
AP-04 Literature and source review	Iskander	22 Oct 2026	Team checks eight scientific sources and synthesis
AP-05 Systems and technology decisions	Alua	22 Oct 2026	Four-system matrix and justified alternatives reviewed
AP-06 Routing model and test plan	Iskander	23 Oct 2026	Eligibility, cost, heuristic and fixtures agreed
AP-07 Requirements and traceability	Iskander	23 Oct 2026	Must criteria map to tests and components

AP-08 Architecture and data model	Alua	23 Oct 2026	Three diagrams and data relations reviewed
AP-09 Prototype walkthrough	Altynay	23 Oct 2026	Core scenarios and error states demonstrated
AP-10 Supervisor review	Iskander	23 Oct 2026	Actual comments and resulting decisions recorded
AP-11 Peer review and submission	All with Altynay coordinating	24 Oct 2026	Access checked, PRs reviewed, exact tag and presentation prepared

These are proposed deadlines. Four interviews are summarized in section 2; completion of record checks and supervisor review must still be tracked. Do not mark outstanding evidence or review complete before it is available. The CSV in the package includes priority, dependencies, expected result and acceptance criteria.

10.3 Development milestones after submission

Milestone	Proposed timing	Completion gate
M1 Feasibility	Weeks 1–2 after submission	Pilot graph, Android/iOS build and one end-to-end route
M2 Core routing	Weeks 3–4	Four presets, tested Dijkstra/A*, unknown handling and benchmark harness
M3 Reporting and client	Weeks 5–6	Report moderation, map, foreground guidance and error handling
M4 Field and user evaluation	Weeks 7–8	Frozen evaluation data, baseline comparison and usability records
M5 Final integration	Weeks 9–10	Defects addressed, reproducible release and thesis results

This ten-week sequence is a planning baseline, not an official defense calendar. Confirm institutional dates with the supervisor and adjust scope or timing before committing. Dependencies: requirements depend on consultation; graph construction depends on boundary and import; route evaluation depends on independent audit; report-aware integration depends on stable API and lifecycle rules.

10.4 Risk register

Risk	Likelihood / impact	Owner	Trigger and response
Missing kerb or slope data	High / High	Alua	Completeness too low for connected strict routes; shrink core and audit critical crossings
Incorrect graph connectivity	Medium / High	Iskander	Route crosses fence or wrong level; add topology fixtures and field checks
Scope exceeds team capacity	High / High	Iskander	Core milestone slips; defer lifts and advanced navigation
Stale or false reports	Medium / High	Alua	Conflicting evidence; moderation, provenance and recheck queue
No iOS build/test access	Medium / High	Altynay	Week 1 build unavailable; arrange access early and report unresolved platform limitation
Limited follow-up and usability recruitment	Medium / High	Altynay	Initial four interviews completed; arrange follow-up clarification and later task-based testing; disclose missing validation
Seasonal changes	High / Medium	Alua	Field conditions differ; date snapshots and limit conclusions
GPS inaccuracy	Medium / Medium	Altynay	Position jumps or low accuracy; manual origin and uncertainty warning
Provider or deployment constraints	Medium / Medium	Alua	Tile/API terms or resources unsuitable; verify provider and keep configurable endpoint
Sensitive report photos	Medium / Medium	Alua	Personal details visible; restrict access, remove EXIF and moderate images

11. Supervisor feedback

According to the team, Islam Omirzak approved the overall topic and a small change was made to its wording. The exact earlier and approved wording, date and written confirmation were not provided. This records the team's account of topic approval; it does not imply that the supervisor has reviewed this package.

Item	Current record	Required next evidence
Topic approval	Reported by the team	Confirm approved title and approval date
Wording revision	Minor revision reported; exact text unavailable	Record previous and approved wording
Scope and pilot	Prepared for review	Comments on polygon, feature boundaries and feasibility
Requirements and evaluation	Prepared for review	Comments on acceptance targets, baseline and evidence
Architecture and prototype	Prepared for review	Comments and links to revisions

At the review, record the material version, comment, decision, owner and resulting PR or document change. If a suggestion is not adopted, explain why. Do not fabricate comments or mark a review complete merely because the general topic was approved.

12. Submission readiness

Component	Material in this package	Remaining work before claiming completion
1 Proposal	Title, team, aim, objectives and four agreed street boundaries	Confirm final title; encode and check GeoJSON polygon
2 Relevance	Four dated interviews, translated summaries and evidence-to-requirement decisions	Retain original notes, consent record and actual guide; clarify participant checking
3 Literature	Eight scientific publications, supporting sources, analysis and synthesis	Team review and incorporate relevant full-text detail where needed
4 Systems	Four systems and eight criteria with primary sources	Optional local hands-on checks; retain U where unverified
5 Technologies	Four decisions with 2–3 alternatives each	Confirm team capability through integration spike
6 Requirements	Interview-informed priorities, acceptance criteria and test mapping	Supervisor review and later task-based user evaluation
7 Architecture	Three editable diagrams and descriptions	Peer review and boundary/data feasibility confirmation
8 UX	Eight static mockup screens and scenarios	Team walkthrough; interactive Figma link optional

9 Evaluation	Tests, metrics, conditions and baselines	Supervisor review; execute after implementation
10 Management	README, tasks and workflow; team reports YouTrack setup	Add actual URLs/IDs, repository contributions and reviews
11 Feedback	Honest topic-approval record and review form	Obtain substantive comments and document improvements

References

[R1] Majic, I., Gegenleithner, T., & Scholz, J. (2026). OpenStreetMap Suitability Analysis for Wheelchair Routing. AGILE GIScience Series, 7, 12. https://doi.org/10.5194/agile-giss-7-12-2026

[R2] Nguyen, T., Islam, M. F., Banerjee, R., Noyce, H. M., Olejniczak, E. M., Smith, R. O., Gani, M. O., & Raychoudhury, V. (2024). MyPath: Accessible Route Generation Using Crowd-Sensed Surface Information. Mobile and Ubiquitous Systems: Computing, Networking and Services, MobiQuitous 2023, Part II. https://doi.org/10.1007/978-3-031-63992-0_3

[R3] Darvishy, A., Hutter, H.-P., & Mosimann, R. (2022). Towards personalized accessible routing for people with mobility impairments. Computers Helping People with Special Needs, 215–220. https://doi.org/10.1007/978-3-031-08648-9_25

[R4] Darko, J., Folsom, L., Pugh, N., Park, H., Shirzad, K., Owens, J., & Miller, A. (2022). Adaptive personalized routing for vulnerable road users. IET Intelligent Transport Systems, 16(8), 1011–1025. https://doi.org/10.1049/itr2.12191

[R5] Labbé, D., Eisenberg, Y., Snyder, D., Shanley, J., Hammel, J. M., & Froehlich, J. E. (2023). Multiple-Stakeholder Perspectives on Accessibility Data and the Use of Socio-Technical Tools to Improve Sidewalk Accessibility. Disabilities, 3, 621–638. https://doi.org/10.3390/disabilities3040040

[R6] Askari, S., Snyder, D., Li, C., Saugstad, M., Froehlich, J. E., & Eisenberg, Y. (2025). Validating Pedestrian Infrastructure Data: How Well Do Street-View Imagery Audits Compare to Government Field Data? Urban Science, 9(4), 130. https://doi.org/10.3390/urbansci9040130

[R7] Li, C., Pang, R., Labbé, D., Eisenberg, Y., Hosseini, M., & Froehlich, J. E. (2025). Accessibility for Whom? Perceptions of Mobility Barriers Across Disability Groups and Implications for Designing Personalized Maps. Proceedings of CHI 2025. https://doi.org/10.1145/3706598.3713421 Author version: https://arxiv.org/abs/2502.19888

[R8] Saha, M., Saugstad, M., Maddali, H. T., Zeng, A., Holland, R., Bower, S., Dash, A., Chen, S., Li, A., Hara, K., & Froehlich, J. (2019). Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data at Scale. Proceedings of CHI 2019, Paper 62. https://doi.org/10.1145/3290605.3300292

Official documentation below was checked on 20 October 2026. It supports design and comparison claims and is not counted as scientific publications.

[R9] Google. (2023). 8 ways we are making daily tasks more accessible. https://blog.google/company-news/outreach-and-initiatives/accessibility/making-daily-tasks-accessible/

[R10] 2GIS. How to build a walking route. https://help.2gis.ru/question/kak-postroit-peshiy-marshrut

[R11] Wheelmap. Mapping. https://news.wheelmap.org/en/mapping/

[R12] openrouteservice. Routing options. https://giscience.github.io/openrouteservice/api-reference/endpoints/directions/routing-options

[R13] OpenStreetMap contributors. Key incline. https://wiki.openstreetmap.org/wiki/Key:incline

[R14] OpenStreetMap contributors. Key kerb. https://wiki.openstreetmap.org/wiki/Key:kerb

[R15] W3C. (2024). Web Content Accessibility Guidelines 2.2. https://www.w3.org/TR/WCAG22/

[R16] Flutter. Accessibility. https://docs.flutter.dev/ui/accessibility

[R17] FastAPI. Features. https://fastapi.tiangolo.com/features/

[R18] NestJS. Documentation. https://docs.nestjs.com/

[R19] Spring. Spring Boot. https://spring.io/projects/spring-boot/

[R20] PostGIS. ST_DWithin. https://postgis.net/docs/ST_DWithin.html

[R21] SpatiaLite. Project documentation. https://www.gaia-gis.it/fossil/libspatialite/index

[R22] MongoDB. Geospatial queries. https://www.mongodb.com/docs/manual/geospatial-queries/

[R23] React Native. Accessibility. https://reactnative.dev/docs/accessibility

[R24] flutter_map. Documentation. https://docs.fleaflet.dev/

[R25] gRPC. Introduction to gRPC. https://grpc.io/docs/what-is-grpc/introduction/